

RAFed: Responsive Augmentation and Approximate Update Method for Federated Learning with Non-IID Data

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Abstract

Federated learning is a distributed collaborative training framework that enables multiple clients to share model updates and jointly train deep neural networks without exchanging raw data. Although extensive research has explored data augmentation techniques in federated settings, the naturally non-IID data distributions among clients render blind augmentation prone to severe degradation of the learned model. To solve this problem, we suggest Responsive Augmentation and Approximate Update Method for Federated Learning with Non-IID Data (RAFed), aimed at alleviating feature shift in client samples. We leverage a Responsive Augmentation Method to accumulate shared data augmentation policy knowledge through local learning, guiding the policy gradient to consider the impact of data augmentation on unseen local data, and employ an Approximate Update Mechanism to reduce communication costs and achieve efficient policy search. To improve the adaptability of data augmentation policies to local data distributions, we introduce a Dynamic Adaptive Method for searching personalized augmentation policies tailored to heterogeneous clients. Experiments on four popular datasets show that RAFed achieves superior test accuracy and lower communication costs compared to related baselines while providing privacy advantages. The code is available via <https://github.com/anonymously123-stcak/RAFed>.

CCS Concepts

• **Computing methodologies** → **Distributed algorithms.**

Keywords

Data Augmentation; Data-free; Heterogeneous Systems; Heterogeneous Federated Learning

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1 Introduction

Federated learning, as a distributed collaborative training framework, enables multiple participants to jointly build models by performing local optimization and global parameter aggregation without sharing raw data [13, 14, 27]. Its core process involves local iterative updates, server-side gradient aggregation, and periodic model broadcasting, aiming to construct a unified global model. However, the non-independent and identically distributed nature of client data often leads to insufficient local representation [3, 9, 45]. The absence of critical feature samples on individual clients results in incomplete class coverage, making it difficult for local models to fit the global data distribution effectively and causing convergence trajectories to deviate from the theoretical optimum.

As a countermeasure, federated data augmentation methods have attracted increasing attention in recent years [12, 17, 50]. These methods aim to enrich local data representations by generating, synthesizing, or reconstructing pseudo-samples, thereby enhancing the generalization capability of federated models. Existing studies can be broadly categorized into two types: collaborative augmentation [42, 55] and personalized augmentation [8, 56]. Collaborative augmentation focuses on information collaboration and sharing across clients, leveraging global consistency mechanisms to improve the diversity and representativeness of augmented data. In contrast, personalized augmentation emphasizes the local adaptability of augmentation policies, incorporating personalized modeling mechanisms to better accommodate non-IID data environments.

Unfortunately, federated data augmentation methods still face severe challenges when client data exhibit significant distributional differences [32, 37, 51]. Although existing approaches attempt to alleviate such discrepancies through global data generation or local policy adjustment, they essentially adopt a static augmentation paradigm, where predefined augmentation rules are uniformly applied to all clients [2, 4]. This paradigm implicitly assumes potential alignability across client data distributions. However, in extreme heterogeneous scenarios such as zero-overlap in label distributions or orthogonal feature spaces, blind augmentation can cause serious model degradation [11, 19, 36, 39]. Specifically, globally generated samples may not align with local feature subspaces, resulting in significant feature shift between the augmented data and the true local distribution. In addition, locally adaptive augmentation policies may overly rely on local statistics, further intensifying gradient conflicts among clients [46, 57].

Federated data augmentation methods would be highly attractive if they could enhance local data representations without introducing feature shift in client samples. To this end, we propose a Responsive Augmentation and Approximate Update Method for

Federated Learning with Non-IID Data (RAFed). The key idea is to share data augmentation policies during training rather than impose a one-way global augmentation rule. RAFed treats local augmentation policies of each client as dynamic knowledge sources and introduces a Responsive Augmentation Method to enable continuous optimization of augmentation policies and the accumulation of local knowledge. Furthermore, an Approximate Update Mechanism is introduced. Specifically, during each local update, the gradient of the global validation loss with respect to augmentation parameters is approximated by the inner product between local model update gradients and validation gradients via first-order Taylor expansion. This enables local computation of policy gradients and eliminates the need for extra inter-client communication. To suppress approximation errors and preserve privacy, a regularization term is incorporated into the inner product computation, and both the augmentation and validation gradients are clipped to ensure optimization stability and security. To accommodate the personalized requirements of heterogeneous clients, we design a Dynamic Adaptive Method that parameterizes the augmentation policy as a neural network. Within a fixed number of local steps, the policy parameters are updated through weighted accumulation based on the inner product of validation and augmentation gradients, thereby enabling multiple rounds of policy optimization within a single communication round.

We elaborate and emphasize three key advantages of RAFed: First, the Responsive Augmentation Method accumulates and shares augmentation policy knowledge through local learning, enabling the policy gradient to explicitly account for the impact of augmentation on previously unseen local samples. Second, the Approximate Update Mechanism based on first-order Taylor expansion allows the policy gradient to be computed locally in parallel without requiring additional exchange of validation gradients. Third, the Dynamic Adaptive Method integrates both global and local information to effectively search for personalized augmentation policies tailored to the data distribution of heterogeneous clients, balancing global consistency and local adaptability. Overall, our main contributions can be summarized as follows:

- We propose a novel federated data augmentation framework named RAFed to mitigate feature shift in client samples under non-IID settings.
- We design a Responsive Augmentation Method to locally accumulate and share policy knowledge, considering its impact on unseen samples. To tackle communication and privacy challenges, we introduce an Approximate Update Mechanism for local policy gradient computation, without exchanging validation gradients across clients.
- We propose a Dynamic Adaptive Method that models the augmentation policy as a neural network, updating it through weighted accumulation based on the inner product of augmentation and validation gradients.
- Experiments on four datasets demonstrate that RAFed achieves superior test accuracy and lower communication cost while inherently preserving privacy.

2 Related Work

2.1 Federated Learning with Non-IID Data

Federated learning aims to collaboratively train a global model across distributed clients. In practice, the inherent diversity of user-generated data makes non-IID data a prevalent challenge [10, 25, 35]. Most existing methods optimize a weighted average of local losses, but this often leads to degraded global performance under highly non-IID settings [1, 40, 49, 52]. To address this, numerous studies have focused on mitigating client conflicts caused by data heterogeneity. FIST-G2 [20] uses a two-stage optimization framework with a gradient memory-based dynamic momentum mechanism to correct local updates by integrating historical gradients and global information. PFedLAH [6] addresses model divergence through personalized learning with feature-adaptive optimization and local regularization, while applying a weight-aware foresight-adaptive client selection strategy. FedAAW [28] minimizes the convergence upper bound to derive optimal client weights and adjusts them adaptively to counter heterogeneity-induced degradation. ACAA [54] introduces an adaptive computation allocation scheme that balances training loads based on client resources, enabling near-synchronous updates. AFed [5] trains a conditional generator on the server to synthesize balanced samples and applies conditional adversarial networks on clients to reduce group bias. FedBDA [47] combines variational inference and stochastic dropout through locally evaluable Bayesian dropout, and employs a global variational attention mechanism to build a joint attention strategy approximating the local-global median distribution for distribution-aware fusion. However, these methods often incur high communication overhead due to the transmission of redundant parameters and may suffer from degraded performance under data sparsity in highly non-IID environments.

2.2 Federated Data Augmentation

Federated learning faces challenges such as degraded convergence and model overfitting due to data heterogeneity and sparsity, stemming from variations in client behaviors and preferences [1, 38, 41]. To address these issues, federated data augmentation methods aim to improve model generalization under non-IID conditions [26, 29, 43]. SlaugFL [30] utilizes a server-side stable diffusion model to generate candidate samples, trains a domain-adaptive GAN with selected local prototypes, and applies dual optimization on the client side through IID data calibration and global prototype refinement. EaDA [53] tackles environmental bias and representation heterogeneity using a dual-module framework that combines prototype learning with counterfactual sample generation. Cycle-Fed [44] improves data diversity through a uniform dropout-based augmentation module and filters reliable pseudo-labels via a dual-reliability cyclic mechanism and discriminator verification. FedPLL [48] uses locally adaptive MixUp to generate synthetic samples and dynamically calibrates soft labels to mitigate noise. RAHFL [15] enhances robust feature representation through hybrid augmentation and allows clients to perform selective one-way learning during collaboration. While these approaches yield performance gains, they often neglect the issue of sample distribution shift caused by indiscriminate augmentation, which can amplify model bias and hinder cross-client generalization in heterogeneous federated scenarios.

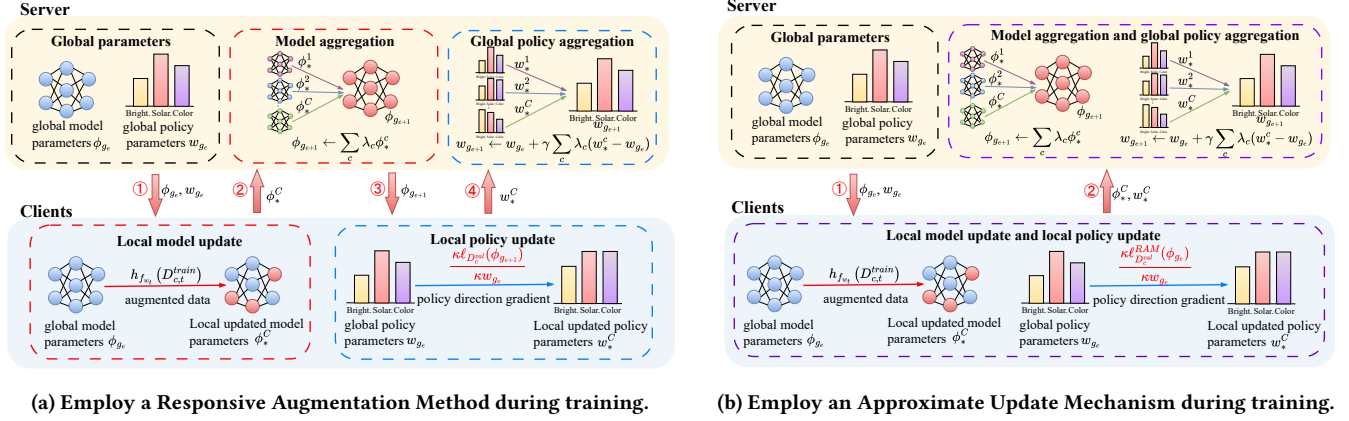


Figure 1: Overall architecture of RAFed: (a) The server sends global model parameters ϕ_g and policy parameters w_g to clients, which train local models using augmented data. The server aggregates updates to compute ϕ_{g+1} , and clients update the policy on ϕ_{g+1} using validation data, followed by a server policy update. (b) Clients aggregate updates via an Approximate Update Mechanism in the Responsive Augmentation Method to form new global model parameters ϕ_{g+1} and policy parameters w_{g+1} .

3 Methodology

To mitigate feature shift in client samples after data augmentation in federated settings, we propose RAFed. Operating under a centralized federated learning framework, RAFed enhances local data via a Responsive Augmentation Method that shares augmentation policies across clients. Unlike conventional policy search methods that optimize augmentation on a single mini-batch, RAFed promotes federation-wide adaptation. Fig. 1 illustrates the model framework, with detailed component descriptions and the overall algorithm provided below.

3.1 Problem Statement

This study builds on FedAvg [33] and focuses on optimizing a data augmentation policy to minimize a specified objective function, as follows:

$$\text{Model} : \min_{\phi} \sum_{c=1}^C \lambda_c \ell(\phi; h_{f_w}(D_c^{train})), \quad (1)$$

$$\text{Policy} : \min_{f_w} \sum_{c=1}^C \lambda_c \ell(\phi; D_c^{val}), \quad (2)$$

where C be the number of clients training the global model ϕ , each with local data D_c and sample count n_c . The augmentation policy f_w generates the transformed dataset $h_{f_w}(D_c^{train})$, with client weights $\lambda_c = \frac{n_c}{n}$, where $n = \sum_c n_c$ and $\sum \lambda_c = 1$. In the global policy setting, all clients share f_w , whereas the local policy setting assigns each client a distinct policy $f_w^{\text{local}} = \{f_w^{\text{local}}, \dots, f_w^{\text{local}}\}$.

We adopt a data augmentation space composed of two sequential operations, following prior work on automated policy search [7, 21, 23, 31]. This study explores 17 augmentation operations: Sharpness, ShearX/Y, Equalize, Rotate, AutoContrast, Brightness, Identity, RandCutout, TranslateX/Y, Solarize, Posterize, Contrast, Color, RandFlip, RandCrop. Each operation's magnitude is rescaled to the normalized interval $[0, 1]$ and randomly sampled from a

uniform distribution. Assuming each of the C clients has an independent policy, the search space includes C joint distributions of size 17×17 . For a global policy, a single 17×17 joint distribution is learned for the two-step augmentation. Specifically, f_w lies in $[0, 1]$, computed via a sigmoid function on parameter w . During sampling, f_w is normalized to sum to one, and samples are drawn from the joint distribution with a regularization term δ . The operation pair (r_1, r_2) is then sampled from this mixed distribution, as follows:

$$(r_1, r_2) \sim (1 - \delta) \cdot \frac{f_w}{\sum(f_w)} + \delta \cdot \frac{1}{289}. \quad (3)$$

This augmentation sampling uses a uniform distribution over all operation pairs to balance exploration and exploitation.

3.2 Responsive Augmentation Method

To enable differentiable augmentation sampling, we adopt inner and outer loops. The model is trained for t steps on training data via stochastic gradient descent, constituting the inner loop, followed by a validation step on unseen validation data, forming the outer loop [16]. Training occurs on augmented data, while evaluation uses validation data, where the loss reflects the augmentation policy's impact on generalization. However, federated learning requires training local models and aggregating a global model, adding complexity. Directly applying the inner-outer loop concept to federated learning is challenging, so we redefine these steps. In standard federated learning, the server sends the global model to clients, who train it locally, and the server consolidates changes to enhance the global model. This local training and global aggregation cycle is treated as an inner loop step. After e such updates, validating the global model on each client constitutes an outer loop step. In our setup, $e = 1$, meaning validation occurs after each training round.

During a single round of client updates, local training is performed on each client, followed by aggregation of local models to update the global weights, which are redistributed to clients for computing validation loss. Each round starts with client c setting its model weight ϕ_0^c equal to the global model ϕ_g . Local training

involves T iterations with a batch size of M for the training data $D_{c,t}^{train}$. In each iteration t , data samples are transformed by the policy $f_{w_t^c}$, and the local loss for client c at iteration t is calculated as follows:

$$L_{local} = \frac{1}{M} \sum_{k=1}^M F_{w_t^c}^k \ell(\phi_t^c; h_{f_{w_t^c}^k}^k(D_{c,t}^{train(k)})), \quad (4)$$

where the sampled transformation $h_{f_{w_t^c}^k}^k$ corresponds to the operation pair (r_1, r_2) , and $F_{w_t^c}^k$ represents the unnormalized probability $f_{(w_t^c)}(r_1, r_2)$ for the k -th transformation. After local updates using the augmentation policy $f_{w_t^c}$, the results are aggregated to produce the new global weight $\phi_{g_{e+1}}$, which is redistributed to the clients for validation loss evaluation. At the start of the e -th training round, the server distributes the global weight ϕ_{g_e} to all clients. The Responsive Augmentation Method (RAM) proceeds as follows: Each client c performs local updates by optimizing its local weight ϕ_t^c with augmented data, using the local loss from Equation (3). The updated local weights are aggregated as $\phi_{g_{e+1}} = \sum_c \lambda_c \phi_t^c$. The server then redistributes $\phi_{g_{e+1}}$ to clients, where the responsive augmentation loss is computed on the validation set, as follows:

$$RAM = \ell_{D_{c,eval}}(\phi_{g_{e+1}}). \quad (5)$$

The data augmentation policy f_w is updated based on gradients calculated via backpropagation. However, this requires additional updates on the same clients, increasing communication costs. Furthermore, it relies on validation gradient information from other clients, potentially raising privacy concerns.

3.3 Approximate Update Mechanism

To minimize communication costs during policy search and prevent privacy leakage by avoiding access to gradients from other clients, we approximate the gradient at local step t on client c with respect to w_{t-1}^c using Taylor expansion. Theorem 3.1 is formulated as follows:

THEOREM 3.1. Given the local weight ϕ_t^c of client c and the global weight ϕ_{g_e} , the loss derived from the updated weight ϕ_t^c at step t on client c , using first-order Taylor expansion, as follows:

$$\ell_{D_{c,eval}}(\phi_{g_{e+1}}) = \ell_{D_{c,eval}}(\phi_t^c) + \nabla \ell_{D_{c,eval}}(\phi_t^c)^T (\phi_{g_e} - \phi_t^c). \quad (6)$$

When computing the loss gradient with respect to w_{t-1}^c , the first-order gradient approximation is given by:

$$-\lambda_c \cdot \beta \frac{\kappa(\nabla \ell_{D_{c,eval}}(\phi_t^c)^T \nabla \ell_{h_{f_{w_{t-1}^c}}(D_{c,t}^{train})}(\phi_{t-1}^c))}{\kappa w_{t-1}^c}, \quad (7)$$

where $\phi_t^c = \phi_{g_e} - \beta \cdot x_{\phi_{g_e}}^{aug} - \dots - \beta \cdot x_{\phi_{t-1}^c}^{aug}$, and $x_{\phi_t^c}^{aug} = \nabla \ell_{h_{f_{w_t^c}}(D_{c,t}^{train})}(\phi_t^c)$.

PROOF. Based on Taylor's theorem, we expand the loss function:

$$\begin{aligned} \frac{\kappa \ell_{D_{c,eval}}(\phi_{g_{e+1}})}{\kappa w_{t-1}^c} &= \frac{\kappa \ell_{D_{c,eval}}(\phi_t^c)}{\kappa w_{t-1}^c} + \frac{\kappa \nabla \ell_{D_{c,eval}}(\phi_t^c)^T (\phi_{g_{e+1}} - \phi_t^c)}{\kappa w_{t-1}^c} \\ &= \nabla \ell_{D_{c,eval}}(\phi_t^c)^T \frac{\kappa \phi_t^c}{\kappa w_{t-1}^c} + \nabla \ell_{D_{c,eval}}(\phi_t^c)^T \frac{\kappa(\lambda_c - 1) \phi_t^c}{\kappa w_{t-1}^c}. \end{aligned} \quad (8)$$

Since the local model update follows:

$$\phi_t^c = \phi_{g_e} - \beta \sum_{\tau=0}^{t-1} x_{\phi_{\tau}^c}^{aug}, \quad (9)$$

substituting this into the expansion yields:

$$\begin{aligned} \frac{\kappa \ell_{D_{c,eval}}(\phi_{g_{e+1}})}{\kappa w_{t-1}^c} &= -\beta \cdot \nabla \ell_{D_{c,eval}}(\phi_t^c)^T \frac{\kappa x_{\phi_{t-1}^c}^{aug}}{\kappa w_{t-1}^c} \\ &\quad - \beta \cdot (\lambda_c - 1) \nabla \ell_{D_{c,eval}}(\phi_t^c)^T \frac{\kappa x_{\phi_{t-1}^c}^{aug}}{\kappa w_{t-1}^c} \\ &= -\lambda_c \cdot \beta \frac{\kappa(\nabla \ell_{D_{c,eval}}(\phi_t^c)^T \nabla \ell_{h_{f_{w_{t-1}^c}}(D_{c,t}^{train})}(\phi_{t-1}^c))}{\kappa w_{t-1}^c}. \end{aligned} \quad (10)$$

The original Responsive Augmentation Method requires clients to compute gradients on the validation set and transmit them to the server for global aggregation. The Approximate Update Mechanism replaces this by directly approximating the global weight $\phi_{g_{e+1}}$ using the locally updated model weight ϕ_t^c , eliminating additional gradient communication. To optimize the augmentation policy, we maximize the inner product between gradients from sampled validation data and augmented data using $f_{w_{t-1}^c}$. Validation data are replaced by the subsequent mini-batch, and gradient computations are performed within the same client to prevent privacy leakage. Theorem 1 shows the loss gradient can be computed concurrently with local updates. However, the inner product in Eq. (7) is sensitive to vector magnitudes, favoring larger gradient norms and introducing bias. To mitigate this, we apply gradient clipping, independently clipping gradients from both validation and augmented data in Eq. (7), regulated by a hyperparameter ϵ .

3.4 Dynamic Adaptive Method

The Approximate Update Mechanism computes the policy gradient at each local update step t without aggregation, averaging these gradients across steps on each client to update the policy after one communication round. However, this slows convergence toward an optimal augmentation policy, as only one gradient descent update occurs per round. To enable faster adaptation and task personalization, we propose a dynamic adaptation method where the augmentation policy is represented as $f_w = \text{sigmoid}([w_1, w_2, \dots, w_{289}])$, with $[w_1, w_2, \dots, w_{289}]$ forming a 17×17 joint distribution. The policy is modeled by a neural network with one embedding layer, two 100-dimensional hidden layers, and a 17×17 output layer. Training the policy involves maximizing the inner product between policy gradients on each client, as formalized in Theorem 2:

THEOREM 3.2. Given the global policy parameter w_{g_e} and the initial policy parameter w_0^c on the i -th validation batch of client c , the updated global policy parameter $w_{g_{e+1}}$ can be expressed as:

$$w_{g_{e+1}} = w_{g_e} - \gamma \mu \frac{\kappa}{\kappa w_0^c} \mathbb{E} \left[\sum_{i=0}^t L_{c,i} - \frac{\mu}{2} \sum_{i=0}^t \sum_{j=0}^{i-1} \langle \nabla L_{c,i}, \nabla L_{c,j} \rangle \right], \quad (11)$$

where $\nabla L_{c,i} = \ell_{D_{c,i}^{val}}^{RAM}(w_0^c)$ represents the responsive augmentation loss in Eq. (5), computed on the i -th validation batch of client c using the global policy parameter w_0^c . The term $\langle \nabla L_{c,i}, \nabla L_{c,j} \rangle$ denotes the dot product between two policy gradients on client c . A positive dot product indicates that gradients align in similar directions, enabling local updates from multiple clients to collaboratively enhance the global optimization objective, accelerate policy search, and support personalized policy adaptation.

PROOF. We start from the global policy parameter update rule:

$$w_{ge+1} = w_{ge} + \gamma \mathbb{E}[-\mu x_{in}^c], \quad (12)$$

where x_{in}^c represents the accumulated gradient on client c during local updates. Based on the Responsive Augmentation Method loss gradient computation, we expand the expectation of x_{in}^c :

$$\mathbb{E}[x_{in}^c] = \mathbb{E}\left[\sum_{i=0}^t \ell'_{R_i}(w_0^c) - \mu \sum_{i=0}^t \sum_{j=0}^{i-1} \ell''_{R_i}(w_0^c) \cdot \ell'_{R_j}(w_0^c)\right]. \quad (13)$$

Next, by substituting the definition of the responsive augmentation loss $L_{c,i}$, we obtain:

$$\mathbb{E}[x_{in}^c] = \mathbb{E}\left[\sum_{i=0}^t \nabla L_{c,i} - \mu \sum_{i=0}^t \sum_{j=0}^{i-1} \nabla^2 L_{c,i} \cdot \nabla L_{c,j}\right]. \quad (14)$$

To further refine the second-order term $\nabla^2 L_{c,i} \cdot \nabla L_{c,j}$, we leverage the symmetry property of gradient exchange terms (i.e., Hessian approximation), yielding:

$$\mathbb{E}[x_{in}^c] = \mathbb{E}\left[\sum_{i=0}^t \nabla L_{c,i} - \frac{\mu}{2} \sum_{i=0}^t \sum_{j=0}^{i-1} (\nabla^2 L_{c,i} \cdot \nabla L_{c,j} + \nabla L_{c,i} \cdot \nabla^2 L_{c,j})\right]. \quad (15)$$

Since the policy gradient update aims to maximize the inner product of gradients, we reformulate it in terms of the expected dot product of gradients:

$$\mathbb{E}[x_{in}^c] = \mathbb{E}\left[\sum_{i=0}^t \nabla L_{c,i} - \frac{\mu}{2} \sum_{i=0}^t \sum_{j=0}^{i-1} \langle \nabla L_{c,i}, \nabla L_{c,j} \rangle\right]. \quad (16)$$

Finally, substituting this into the global policy update equation and incorporating the regularization term $\frac{\kappa}{\kappa w_0^c}$, we derive the final update formula:

$$w_{ge+1} = w_{ge} - \gamma \mu \frac{\kappa}{\kappa w_0^c} \mathbb{E}\left[\sum_{i=0}^t L_{c,i} - \frac{\mu}{2} \sum_{i=0}^t \sum_{j=0}^{i-1} \langle \nabla L_{c,i}, \nabla L_{c,j} \rangle\right]. \quad (17)$$

3.5 Collaborative Optimisation

A pretrained network can update the policy, but this approach poses challenges in federated learning: pretraining is cumbersome, requires a sequential standalone phase before model training, and lacks synchronization with main model optimization. Without direct feedback from the main model, policy parameters struggle to adapt to the current model state and data distribution. In federated learning with high client data heterogeneity, static policies often lack generalization. The dynamic adaptive approach addresses this by enabling joint optimization of model training and policy update, using weights ϕ_{t+1}^c and gradients at each local update step t to simultaneously update the model and policy.

The collaborative optimization procedure of RAFed is described in Algorithm 1, where the server initializes the global model ϕ_{ge} and policy parameters w_{ge} , then transmits them to all clients. At each local update step t , clients use the current policy f_{w_t} to augment their training data $D_{c,t}^{train}$, yielding $h_{f_{w_t}}(D_{c,t}^{train})$, which updates the model. Clients then compute the gradient $\nabla \ell_{D_{c,t}^{val}}(\phi_{t+1}^c)$ on their validation set $D_{c,t}^{val}$ using updated model parameters ϕ_{t+1}^c . According

Algorithm 1 The Collaborative Optimization Process of RAFed.

Input: Communication round E , number of clients T , global policy learning rate γ , gradient clipping threshold ϵ , local model learning rate ∂ , local steps P , regularization term δ , and local policy learning rate μ .

Output: Updated local parameters ϕ_{t+1}^c , updated global policy parameters w_{t+1}^c .

```

1: Initialize the global model parameters  $\phi_{ge}$  and the global policy
   parameters  $w_{ge}$ ;
2: for  $e = 1, \dots, E$  do
3:   Sample  $C$  clients from  $1, \dots, T$  clients;
4:   for  $c = 1, \dots, C$  do
5:     Set  $\phi_0^c = \phi_{ge}$  and  $w_0^c = w_{ge}$ ;
6:      $\phi_*^c, w_*^c \leftarrow \text{LocalUpdate}(\phi_0^c, w_0^c)$ ;
7:   end for
8:    $\phi_{ge+1} \leftarrow \sum_c \lambda_c \phi_*^c$ ;
9:    $\Delta w \leftarrow \sum_c \lambda_c (w_*^c - w_{ge})$ ;
10:   $w_{ge+1} \leftarrow w_{ge} + \gamma \Delta w$ .
11: end for
12: Procedure LocalUpdate( $\phi, w$ ):
13:   Initialize  $\phi_0^c = \phi$  and  $w_0^c = w$ ;
14:   for  $t = 0, \dots, P$  do
15:     Sample transformations  $h_{f_{w_t}}$  from  $f_{w_t}$ ;
16:     Sample batch data  $h_{f_{w_t}}(D_{c,t}^{train})$  from  $D_c$  and  $h_{f_{w_t}}$ ;
17:     Compute  $\nabla \ell_{h_{f_{w_t}}(D_{c,t}^{train})}(\phi_t^c)$ ;
18:      $\phi_{t+1}^c \leftarrow \phi_t^c - \partial \nabla \ell_{h_{f_{w_t}}(D_{c,t}^{train})}(\phi_t^c)$ ;
19:     Sample batch data  $D_{c,t}^{val}$  from  $D_c$ ;
20:     Compute  $\nabla \ell_{D_{c,t}^{val}}(\phi_{t+1}^c)$  at  $\phi_{t+1}^c$ ;
21:      $w_{t+1}^c \leftarrow w_t^c - \mu \nabla \ell_{D_{c,t}^{val}}^{RAM}(\phi_{t+1}^c)$ ;
22:   end for
23:   Set  $\phi_*^c = \phi_{t+1}^c$  and  $w_*^c = w_{t+1}^c$ ;
24:   return  $\phi_*^c$  and  $w_*^c$ .
    
```

to Theorem 1, the policy parameters w_t^c are updated to maximize the gradient alignment between the validation and the augmented data by increasing the inner product $\nabla \ell_{h_{f_{w_t}}(D_{c,t}^{train})}(\phi_t^c)$. These model and policy updates occur at each local step, followed by server aggregation of clients' parameters using weights λ_c .

Federated learning's limitation is the computational burden of performing policy gradient backpropagation at each step. To reduce this, we adopt a periodic policy update strategy, updating the policy only when $t \bmod m == 0$, with $m = 10$ in all experiments. The policy network is also simplified to two hidden layers with 25 dimensions each.

3.6 Complexity Analysis

3.6.1 Computational Complexity. The complexity of RAFed is analyzed for both the local client and the global server. For a local client, model parameter updates ϕ_t^c require P iterations on local training data with batch size M , resulting in $O(P \times M \times d)$, where d is the model parameter dimension. Updating data augmentation policy parameters w_t^c , with dimension d' , has complexity $O(P \times M \times d')$. Each client experiences a computational complexity of $O(P \times M \times (d + d'))$.

Table 1: The accuracy (% , mean $\pm$ std) of RAFed compared to baseline models on four heterogeneous datasets, where the mean represents the average result over 10 runs. Bold indicates the best performance.

Method	α	FedAvg	FIST-G2	FedAAW	AFed	SlaugFL	Cycle-Fed	FedPLL	RAHFL	RAFed
CIFAR-10	0.05	12.09 $\pm$ 2.03	36.64 $\pm$ 0.75	37.86 $\pm$ 0.82	36.82 $\pm$ 1.64	39.41 $\pm$ 1.38	42.47 $\pm$ 0.58	39.48 $\pm$ 1.13	40.31 $\pm$ 0.79	49.23$\pm$0.38
	0.1	17.12 $\pm$ 0.39	42.69 $\pm$ 0.91	52.14 $\pm$ 0.36	49.24 $\pm$ 1.35	47.45 $\pm$ 1.09	46.53 $\pm$ 0.21	51.37 $\pm$ 0.35	59.52 $\pm$ 1.35	63.02$\pm$0.52
	0.5	29.05 $\pm$ 0.14	45.69 $\pm$ 0.58	61.25 $\pm$ 0.19	53.76 $\pm$ 1.53	55.12 $\pm$ 1.64	63.27 $\pm$ 0.45	60.87 $\pm$ 0.53	68.35 $\pm$ 1.65	72.31$\pm$0.59
	1	35.32 $\pm$ 0.31	55.37 $\pm$ 0.84	65.30 $\pm$ 0.31	59.69 $\pm$ 1.78	60.08 $\pm$ 2.12	62.50 $\pm$ 0.34	63.22 $\pm$ 0.15	71.33 $\pm$ 2.23	80.82$\pm$0.22
CIFAR-100	0.05	4.68 $\pm$ 0.25	16.05 $\pm$ 0.63	20.42 $\pm$ 0.34	15.74 $\pm$ 0.86	17.42 $\pm$ 0.41	21.55 $\pm$ 0.51	19.46 $\pm$ 1.52	21.20 $\pm$ 1.55	24.87$\pm$0.56
	0.1	6.73 $\pm$ 0.28	28.16 $\pm$ 0.44	30.23 $\pm$ 0.25	29.22 $\pm$ 1.32	27.91 $\pm$ 1.25	27.34 $\pm$ 0.46	34.11 $\pm$ 0.38	28.62 $\pm$ 1.46	38.15$\pm$0.51
	0.5	11.54 $\pm$ 0.27	32.34 $\pm$ 0.81	38.64 $\pm$ 0.32	34.27 $\pm$ 0.56	35.90 $\pm$ 1.34	39.12 $\pm$ 0.56	38.42 $\pm$ 0.51	40.29 $\pm$ 1.29	43.92$\pm$0.61
	1	13.31 $\pm$ 0.06	36.41 $\pm$ 0.58	42.83 $\pm$ 0.18	40.32 $\pm$ 1.09	38.96 $\pm$ 1.40	42.52 $\pm$ 0.85	39.73 $\pm$ 0.43	43.56 $\pm$ 1.38	47.67$\pm$0.67
SVHN	0.05	20.42 $\pm$ 2.53	49.24 $\pm$ 0.46	45.27 $\pm$ 0.42	48.23 $\pm$ 1.25	49.63 $\pm$ 1.75	53.35 $\pm$ 1.69	48.39 $\pm$ 1.09	54.34 $\pm$ 1.51	60.75$\pm$0.93
	0.1	36.65 $\pm$ 0.69	50.25 $\pm$ 0.68	57.43 $\pm$ 0.33	53.74 $\pm$ 1.33	53.38 $\pm$ 1.35	58.53 $\pm$ 0.42	56.17 $\pm$ 0.65	63.25 $\pm$ 1.59	67.34$\pm$0.20
	0.5	49.33 $\pm$ 0.18	64.83 $\pm$ 0.41	80.83 $\pm$ 0.39	67.49 $\pm$ 1.58	65.41 $\pm$ 1.76	80.75 $\pm$ 0.18	80.32 $\pm$ 0.62	80.89 $\pm$ 1.93	86.25$\pm$0.11
	1	57.52 $\pm$ 0.80	73.24 $\pm$ 0.55	81.34 $\pm$ 0.35	78.51 $\pm$ 1.92	75.64 $\pm$ 1.79	82.34 $\pm$ 0.26	81.14 $\pm$ 0.36	82.23 $\pm$ 1.14	88.92$\pm$0.19
Tiny-ImageNet	0.05	5.56 $\pm$ 0.54	12.53 $\pm$ 0.37	16.54 $\pm$ 0.57	14.87 $\pm$ 1.86	16.23 $\pm$ 1.42	14.39 $\pm$ 0.76	19.64 $\pm$ 0.72	20.01 $\pm$ 1.32	22.56$\pm$0.23
	0.1	8.29 $\pm$ 0.24	17.42 $\pm$ 0.29	23.35 $\pm$ 0.67	20.01 $\pm$ 1.67	20.05 $\pm$ 1.20	16.27 $\pm$ 0.34	23.36 $\pm$ 0.41	22.86 $\pm$ 1.17	26.54$\pm$0.21
	0.5	13.54 $\pm$ 0.07	18.65 $\pm$ 0.66	24.37 $\pm$ 0.21	27.12 $\pm$ 1.53	24.66 $\pm$ 1.19	18.64 $\pm$ 0.29	29.05 $\pm$ 0.45	30.13 $\pm$ 1.21	32.28$\pm$0.16
	1	13.67 $\pm$ 0.15	28.46 $\pm$ 0.54	29.30 $\pm$ 0.38	30.87 $\pm$ 1.26	28.87 $\pm$ 1.07	29.61 $\pm$ 0.53	33.23 $\pm$ 0.38	31.67 $\pm$ 1.96	36.13$\pm$0.14

The server aggregates parameters from all clients, resulting in a complexity of $O(C \times d)$ for models and $O(C \times d')$ for policies, summing to $O(C \times (d + d'))$.

3.6.2 Communication Complexity. In the RAFed framework, communication complexity arises from data transfer between local clients and the global server. Each round, clients send updated model parameters ϕ_*^c and policy parameters w_*^c to the server, with a transfer complexity of $O(C \times (d + d'))$, where d and d' are the dimensions of model and policy parameters, and C is the number of clients. Similarly, the server sends updated global model parameters $\phi_{g_{e+1}}$ and policy parameters $w_{g_{e+1}}$ back to the clients, with the same complexity $O(C \times (d + d'))$. Thus, the total communication complexity per round is $O(C \times (d + d'))$.

4 Experiments

4.1 Experimental setup

4.1.1 Datasets. We evaluate RAFed and baseline methods on four widely used classification benchmarks: CIFAR-10 and CIFAR-100 [22], SVHN [34], and Tiny-ImageNet [24]. CIFAR-10 and CIFAR-100 consist of 60,000 32 $\times$ 32 RGB images across 10 and 100 categories, respectively. SVHN contains over 600,000 digit images from Google Street View. Tiny-ImageNet, a subset of ImageNet, comprises 100,000 64 $\times$ 64 images spanning 200 categories.

4.1.2 Baselines. We conduct a comprehensive comparison of RAFed with several advanced federated learning approaches, including FedAvg [33], FIST-G2 [20], FedAAW [28], and AFed [5], as well as recent data augmentation methods tailored for federated settings, such as SlaugFL [30], Cycle-Fed [44], FedPLL [48], and RAHFL [15].

4.1.3 Implementation Details. Experiments are performed on an Intel Xeon Gold 6140 CPU and an 8-node GeForce RTX 2080 Ti GPU, with classification accuracy as the evaluation metric. The global model consists of three convolutional layers (64 filters, 3 $\times$ 3

Table 2: The communication overhead required for RAFed to achieve the same accuracy as FedAvg at 100 rounds when $\alpha = 0.05$ on CIFAR-10 and CIFAR-100, compared to baseline models. The units for param and total are MB.

Method	CIFAR-10			CIFAR-100		
	rounds	param	total	rounds	param	total
FedAvg	100(6.3 $\times$)	3.61	361(5.4 $\times$)	100(2.9 $\times$)	1085.5	108550(2.9 $\times$)
FIST-G2	53(3.3 $\times$)	3.61	191.33(2.9 $\times$)	77(2.3 $\times$)	1085.5	83583.5(2.3 $\times$)
FedAAW	29(1.8 $\times$)	7.22	281.58(4.2 $\times$)	63(1.9 $\times$)	2171	136773(3.7 $\times$)
AFed	21(1.3 $\times$)	3.7	77.7(1.2 $\times$)	45(1.3 $\times$)	1086.47	48891.15(1.3 $\times$)
SlaugFL	26(1.6 $\times$)	3.61	93.86(1.4 $\times$)	71(2.1 $\times$)	1085.5	77070.5(2.1 $\times$)
Cycle-Fed	32(2 $\times$)	4.28	136.96(2.1 $\times$)	46(1.4 $\times$)	1104.3	50797.8(1.4 $\times$)
FedPLL	38(2.4 $\times$)	7.22	274.36(4.1 $\times$)	82(2.4 $\times$)	2171	178022(4.8 $\times$)
RAHFL	56(3.5 $\times$)	4.54	254.24(3.8 $\times$)	65(1.9 $\times$)	1135.9	73833.5(2 $\times$)
RAFed	16(1 $\times$)	4.16	66.56(1 $\times$)	34(1 $\times$)	1092.5	37145(1 $\times$)

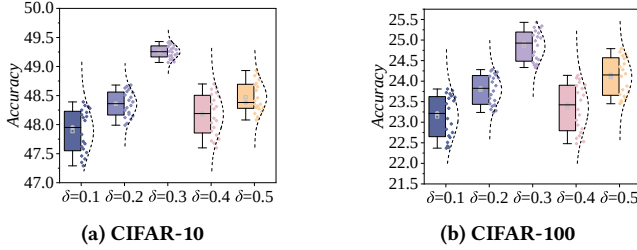
kernels), three fully connected layers (256, 128 units), and a classification layer. To simulate heterogeneity and evaluate robustness on non-IID data, samples are distributed to 100 clients via Dirichlet distributions with $\alpha = \{0.05, 0.1, 0.5, 1\}$, where smaller α denotes greater heterogeneity. Training is conducted for $E = 500$ rounds with server and client policy learning rates $\gamma = 0.1$, $\mu = 0.5$, client learning rate $\partial = 0.1$, regularization $\epsilon = 0.3$, local epochs set to 10, and batch size 128. Baselines adhere to their original settings for fair comparison.

4.2 Comparison Results

4.2.1 Accuracy Comparison. The empirical results summarized in Table 1 confirm the superior performance of RAFed under a range of heterogeneity settings ($\alpha = \{0.05, 0.1, 0.5, 1\}$) and across diverse datasets. The results show that RAFed consistently outperforms baseline methods in heterogeneous environments, achieving notable improvements of 13.30%, 9.44%, 7.99%, and 8.73% over the best

Table 3: The effectiveness of the components in RAFed.

Method	α	w/o RAM	w/o DAM	w/o AUM	RAFed
CIFAR-10	0.05	41.06 $\pm$ 0.24	45.52 $\pm$ 0.22	48.35 $\pm$ 0.39	49.23$\pm$0.38
	0.1	56.55 $\pm$ 0.61	60.87 $\pm$ 0.54	62.35 $\pm$ 0.57	63.02$\pm$0.52
	0.5	63.12 $\pm$ 0.52	69.10 $\pm$ 0.55	71.27 $\pm$ 0.61	72.31$\pm$0.59
	1	75.52 $\pm$ 0.39	78.10 $\pm$ 0.36	79.98 $\pm$ 0.25	80.82$\pm$0.22
CIFAR-100	0.05	15.64 $\pm$ 0.66	19.22 $\pm$ 0.63	23.26 $\pm$ 0.78	24.87$\pm$0.56
	0.1	30.24 $\pm$ 0.61	34.32 $\pm$ 0.57	37.30 $\pm$ 0.52	38.15$\pm$0.51
	0.5	36.99 $\pm$ 0.53	40.27 $\pm$ 0.48	42.56 $\pm$ 0.40	43.92$\pm$0.61
	1	41.23 $\pm$ 0.75	44.21 $\pm$ 0.69	46.35 $\pm$ 0.77	47.67$\pm$0.67


Figure 2: Impact of the regularization term δ in the joint distribution when $\alpha = 0.05$.

baseline on CIFAR-10, CIFAR-100, SVHN, and Tiny-ImageNet, respectively. Even in highly heterogeneous settings ($\alpha = 0.05$), RAFed improves accuracy by more than 10% compared to the best baseline. Federated data augmentation methods, such as Cycle-Fed, FedPLL, and RAHFL, generally outperform traditional optimization-based methods by mitigating performance degradation caused by data heterogeneity, whereas the latter struggle with fully leveraging local data characteristics. RAFed combines responsive augmentation with an approximate update mechanism, enhancing global model generalization while preserving data diversity, resulting in the best performance among all methods. The relative improvement of RAFed varies across datasets, with more pronounced gains on CIFAR-10 and SVHN, showcasing its effectiveness in image classification tasks. On Tiny-ImageNet, while RAFed still outperforms baselines, the improvement is smaller, likely due to the dataset’s complexity and the augmentation method’s adaptability.

4.2.2 Communication Cost Comparison. Table 2 compares the communication cost of various methods in achieving the same accuracy as FedAvg at 100 rounds on CIFAR-10 and CIFAR-100 when $\alpha = 0.05$. The parameter column indicates the amount of data transmitted per round, while the total column is the product of the transmitted data and the required rounds to achieve the target accuracy. Results show that RAFed achieves the lowest communication cost for both CIFAR-10 and CIFAR-100. On CIFAR-10, RAFed reaches the target accuracy in just 16 rounds, with a total cost of 66.56 MB—an 81.55% reduction compared to FedAvg. For CIFAR-100, RAFed achieves the target in 34 rounds, with a total cost of 37,145 MB, reducing communication overhead by 65.78%. While baseline methods like FIST-G2 and AFed reduce costs under certain conditions, their communication overhead remains significantly higher than RAFed’s. This

highlights RAFed’s ability to minimize communication costs while maintaining strong model performance, validating its efficiency.

4.3 Ablation Study

To examine the individual impact of RAFed’s components, we design several ablation variants: w/o RAM removes the Responsive Augmentation Method, w/o DAM eliminates the Dynamic Adaptive Method, and w/o AUM omits the Approximate Update Mechanism. As shown in Table 3, RAFed consistently achieves superior performance across all evaluation settings, underscoring the necessity and complementary benefits of these components. The most significant performance degradation occurs when the Responsive Augmentation Method is removed, highlighting its pivotal role in mitigating feature shift and ensuring robustness across heterogeneous client distributions. Similarly, the omission of the Approximate Update Mechanism leads to a pronounced drop in accuracy under high heterogeneity with $\alpha = 0.05$, demonstrating its effectiveness in maintaining model stability and convergence efficiency. Furthermore, removing the Dynamic Adaptive Method weakens the augmentation policy’s capacity to adaptively align with local data characteristics, thereby constraining the model’s ability to learn discriminative representations in non-IID environments.

4.4 In-depth Analysis

4.4.1 Analysis of the Regularization Term δ . When $\alpha = 0.05$, Fig. 2 illustrates the impact of the regularization term δ in the joint distribution. The model achieves the best performance on both CIFAR-10 and CIFAR-100 when $\delta = 0.3$, suggesting that this level of regularization effectively balances the trade-off between model flexibility and stability under data augmentation. In particular, a moderate δ value mitigates overfitting to augmented samples while preserving sufficient diversity in the learned representations, thereby enhancing the model’s generalization capability across datasets.

4.4.2 Scalability Analysis. We assess RAFed’s scalability by varying the number of participating clients $C = \{5, 10, 30, 50, 100\}$ on a fixed dataset and evaluating the resulting test accuracy across different methods. As shown in Table 4, overall performance gradually declines as the number of clients increases, primarily due to higher communication overhead, increased data heterogeneity, and the growing difficulty of achieving consistent global aggregation. Nevertheless, RAFed consistently surpasses all baseline methods under every setting, demonstrating its strong robustness and adaptability in large-scale federated environments. Notably, even when the number of clients reaches 100, RAFed maintains competitive accuracy with only a marginal performance drop, highlighting its ability to effectively alleviate degradation caused by client expansion.

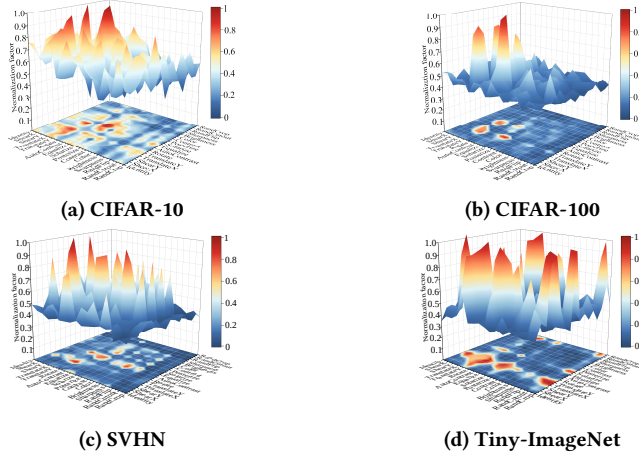
4.4.3 Extreme Label Skew Analysis. Table 5 presents the experimental results under varying levels of label skew, where Z denotes the number of label categories assigned to each client. As expected, in extreme skew scenarios such as $Z = 1$, where each client possesses data from only a single class, the performance of all models declines sharply due to the severe limitation of local label diversity and the resulting distributional discrepancy across clients. Nevertheless, RAFed consistently achieves superior performance compared with all baseline methods, highlighting its strong capacity to cope with

Table 4: The accuracy of RAFE and baseline models with different numbers of clients on CIFAR-10 when $\alpha = 0.05$.

C	FedAvg	FIST-G2	FedAAW	AFed	SlaugFL	Cycle-Fed	FedPLL	RAHFL	RAFed
5	20.10±0.92	47.78±0.58	55.36±0.65	53.44±1.53	53.72±0.93	51.26±0.62	47.89±0.99	49.35±0.60	62.50±0.23
10	16.48±1.42	43.32±0.62	49.60±0.47	47.54±1.26	51.34±1.22	48.54±0.54	45.56±1.25	46.88±0.53	57.34±0.25
30	15.14±1.35	41.84±0.91	44.15±0.59	41.02±1.44	45.78±1.47	46.42±0.77	44.37±0.76	45.59±0.65	54.95±0.31
50	14.20±1.64	38.12±0.66	41.73±0.78	39.76±1.58	41.76±1.34	43.29±0.49	43.01±1.02	42.06±0.71	51.80±0.34
100	12.09±2.03	36.64±0.75	37.86±0.82	36.82±1.64	39.41±1.38	42.47±0.58	39.48±1.13	40.31±0.79	49.23±0.38

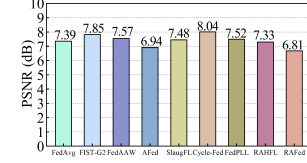
Table 5: The accuracy of RAFE and baseline models on CIFAR-10 under the quantity-based label skew setting when $\alpha = 0.05$.

Z	FedAvg	FIST-G2	FedAAW	AFed	SlaugFL	Cycle-Fed	FedPLL	RAHFL	RAFed
1	3.78±2.92	7.29±1.74	6.33±1.80	6.12±2.53	7.05±1.98	7.72±1.61	7.18±2.04	7.35±1.52	9.32±0.91
2	8.04±2.27	33.42±1.06	29.35±1.23	27.36±1.97	32.18±1.73	35.23±1.02	31.42±1.39	32.65±1.16	41.95±0.54
3	10.13±2.15	35.71±0.82	32.67±1.04	28.93±1.84	34.37±1.52	36.24±0.73	35.23±1.30	34.81±0.94	44.16±0.47

**Figure 3: Visualization of the global data augmentation policies when $\alpha = 0.05$.**

pronounced label imbalance and mitigate the adverse effects of non-IID label distributions. As Z increases, the overall performance of all methods improves, reflecting that more diverse local label distributions contribute to better representation learning and enhanced global model generalization.

4.4.4 Visualization Analysis. Fig 3 illustrates the global data augmentation policies learned by the Responsive Augmentation Method across different datasets. The results show that the learned policies adaptively generate augmentation operations based on dataset characteristics, effectively mitigating feature shifts in client samples. The augmentation policies vary significantly across datasets: on CIFAR-100, operations like Solarize and TranslateX enhance feature representation in complex backgrounds; on SVHN, geometric transformations like rotation and scaling accommodate street-view digit variations; on Tiny-ImageNet, morphological transformations improve adaptability to complex scenes and diverse object categories.

**Figure 4: PSNR (dB) values for privacy evaluation when $\alpha = 0.05$ on CIFAR-10.**

4.4.5 Privacy Protection Analysis. We evaluate the privacy-preserving capability of RAFE using the representative baseline of Peak Signal-to-Noise Ratio (PSNR) [18]. For both RAFE and baseline federated learning methods, we perform DLG attacks on the entire model. Lower PSNR values indicate better privacy protection. As shown in Fig. 4, the PSNR values corresponding to RAFE are consistently lower than those of other baselines. This is attributed to RAFE's approximate update mechanism, which computes gradients locally within the same client, effectively preventing cross-client gradient leakage.

5 Conclusion

This paper introduces Responsive Augmentation and Approximate Update Method for Federated Learning with Non-IID Data to address feature shift, reduce communication costs, and improve model performance in Non-IID settings. RAFE combines Responsive Augmentation Method to optimize data augmentation policies and an Approximate Update Mechanism for efficient policy search, mitigating feature shift during federated training. A Dynamic Adaptive Method further enhances policy adaptability to heterogeneous client data. Experiments demonstrate RAFE's superiority in test accuracy across datasets while significantly reducing communication overhead. Future work will explore its use in complex federated scenarios and dynamic data augmentation in online learning for improved scalability.

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